Group Coursework Submission Form

Specialist Masters Programme

Please list all names of group members: (Surname, first name) 1. Evans, Benjamin 2. Basmah Khan 3. Ardi Wira Sudarmo	4. 5. 6. 7. **GROUP NUMBER:** **4**
MSc in: Actuarial Science (/with Business Analytics)	
Module Code: SMM634	
Module Title: Analytics Methods for Business	
Lecturer: Professor Rosalba Radice	**Submission Date: 29/10/2025**

Declaration:

By submitting this work, we declare that this work is entirely our own except those parts duly identified and referenced in my submission. It complies with any specified word limits and the requirements and regulations detailed in the coursework instructions and any other relevant programme and module documentation. In submitting this work we acknowledge that we have read and understood the regulations and code regarding academic misconduct, including that relating to plagiarism, as specified in the Programme Handbook. We also acknowledge that this work will be subject to a variety of checks for academic misconduct.

We acknowledge that work submitted late without a granted extension will be subject to penalties, as outlined in the Programme Handbook. Penalties will be applied for a maximum of five days lateness, after which a mark of zero will be awarded.

Marker's Comments (if not being marked on-line):

Deduction for Late Submission: | | **Final Mark:** | %

SMM634 - Group Assignment (Group 4)

Contents

1 Introduction **2**
1.1 Identifying Our Response Variable . . . 2
1.2 Regressor Selection . . . 3
2 General Linear Regression Model **4**
2.1 Choice of Regression Model . . . 4
2.2 Assumptions underlying the model . . . 5
3 Results **6**
4 Limitations **7**
5 Future Improvements **9**
Appendices **10**
A GitHub Resources **11**
B Supplementary Figures and Tables **12**
C R Code **14**
C.1 Extracting car manufacturer information . . . 14
C.2 Defining and refining our models . . . 14

1 Introduction

The data given for this assignment appear to be a subset of 1985 Ward's Automotive Yearbook (Schlimmer, 1985). Schlimmer's (1985) dataset contains an additional variable, the average loss per car per year normalised for all cars within a particular size classification (`normalized-losses`). Whilst this variable would be of considerable interest to investigate, particularly for Actuarial Science students, we opted to solely use the `car_price` data for this project.

1.1 Identifying Our Response Variable

The analysis of the dataframe `car_price` for this project was not accompanied by any specific research questions. We decided to make our own to provide a clear and meaningful framework for the interpretation and critical evaluation of our model. Our primary (self-determined) aim for this project was to establish **what properties determine a car's fuel efficiency value**, where we define the fuel efficiency value as the ratio of highway miles per gallon (MPG) to price. A secondary aim was to produce a model to predict the price of a car (in 1985). There are two advantages to this secondary aim. Firstly, even in isolation a predictive model for a car's price is valuable and secondly a price prediction model allows us to decompose our fuel efficiency model into `highwaympg` and `price` (Eq. 1). To assist with the model interpretation, and to ensure we stayed within the scope of the module and assignment, we decided to use the same predictors for both models.

There are two MPG variables in `car_price`: highway (`highwaympg`) and city (`citympg`). Given their strong correlation (Pearson $r = 0.971$), only one was used to reduce noise and variance inflation in coefficient estimates. We chose to use `highwaympg` as steady-state travel on a highway is more consistent and reliable than the MPG obtained during 'stop-and-start' driving patterns in a city.

As shown in Figure 4, there are some non-linear relationships between `highwaympg/price` and variables including `carwidth`, `horsepower` and `enginesize`. Price itself shows a clear right skewed distribution (Figure 1a) and the spread of values widens for higher prices, indicating heteroscedasticity. **To address this we applied a log transformation to `highwaympg/price` and `price` (Figure 1b and 5) and decided to use both as our response variables.** This transformation reduces skew, stabilises the variance and results in clear linear relationships in the plots.

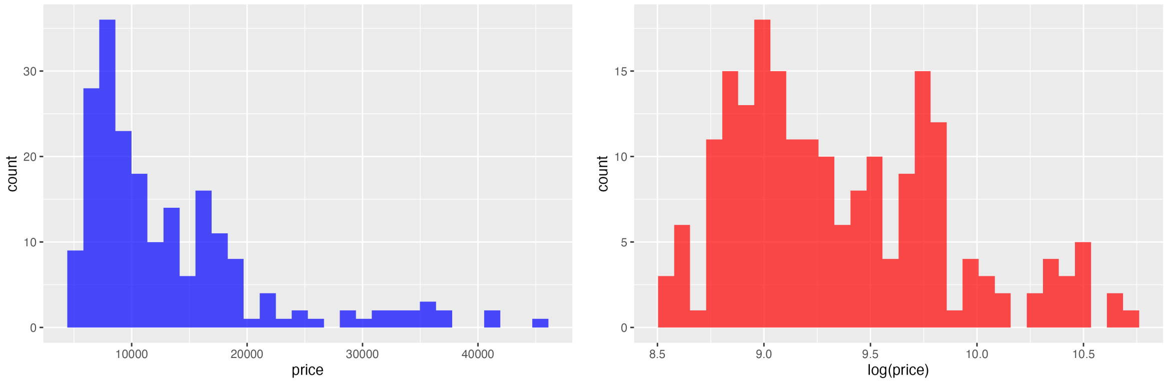

Figure 1: Distributions of car (a) Price and (b) $\log(\text{Price})$. A log transform reduces skew and results in a more symmetric distribution.

The log transformation allows for the response variable to be broken down into two individual components, enabling us to tease out the effects of each explanatory variable on the highwaympg and price separately (as shown below in Eq. 1).

$$\log\left(\frac{\text{highwaympg}}{\text{price}}\right) = \log(\text{highwaympg}) - \log(\text{price}) \tag{1}$$

1.2 Regressor Selection

Table 1: Regressors initially selected for our model, the data type (categorical, numerical (continuous), or numerical (discrete)), along with the inclusion rationale. The categorical data were not broken down further into nominal, ordinal, and binary. *This is a custom factor (see C.1).

Variable	Type	Inclusion rationale
`aspiration`	Categorical	Engine performance
`carbody`	Categorical	Car class and shape
`carheight`	Numeric (continuous)	Car size / weight
`carManufacturer*`	Categorical	Brand and reputation
`compressionratio`	Numeric (continuous)	Engine performance
`curbweight`	Numeric (discrete)	Car size / weight
`cylinderNum`	Categorical	Engine performance
`carwidth`	Numeric (continuous)	Car size / weight
`enginelocation`	Categorical	Car class and shape / Engine performance
`enginetype`	Categorical	Engine performance
`enginesize`	Numeric (discrete)	Engine performance
`fuelsystem`	Categorical	Engine performance
`peakrpm`	Numeric (discrete)	Engine performance
`stroke`	Numeric (continuous)	Engine performance (linked to torque)
`wheelbase`	Numeric (continuous)	Car size (also linked to ride quality)
`horsepower`	Numeric (discrete)	Engine performance

The regressors selected for our model are shown in Table 1. We used prior domain knowledge to select regressors that are likely to affect either the fuel efficiency or the price of a car in 1985.

Some regressors capture a car's size or powertrain specifications which are the primary determinants of both manufacturing cost (influencing `price`) and fuel consumption (influencing `highwaympg`). Other regressors were selected to incorporate both the brand value of different car manufacturers (Figure 2) and the vehicle class, neither of which are fully encapsulated by quantitative data.

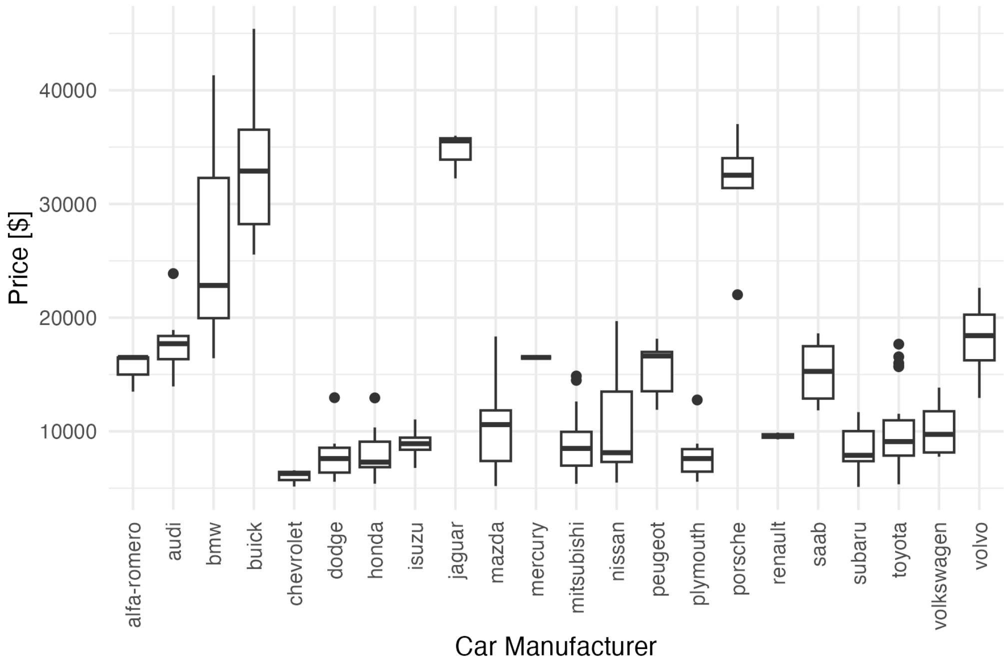

Figure 2: Boxplots showing the variation in price across cars produced by different manufacturers in 1985.

2 General Linear Regression Model

2.1 Choice of Regression Model

A multiple linear regression (MLR) model, implemented in R (see Section C.2), was used to address our research question. This was a natural choice as our response variable is continuous, we wished to model our response variable as a function of multiple quantitative and qualitative predictors and, critically, we assumed a linear relationship between our response variable and predictors.

As outlined in Tables 2 & 3, an iterative mixed stepwise process was used to select the final model (using Akaike's Information Criterion (AIC)), starting from a 'full' model (using all regressors in Table 1) and iterating until no further AIC improvement was obtained. Both models (lmpg2) and (lprice2) were then compared and the regressor sets were manually harmonised.

In ensuring that both models having the same set of final regressors, there is a slight sacrifice on model performance in terms of AIC and adjusted R^2, but this is offset by a gain in model interpretability as we are able to decompose the total effect on $\log(\text{highwaympg}/\text{price})$ into individual components.

Table 2: Variable selection iteration process for $\log(\text{highwaympg}/\text{price})$.

Model	Regressor variables	R^2_{adj}	AIC	Remarks
lmpg1	Selection from domain knowledge	0.9656	-210.7	Initial model
lmpg2	− `cylinderNum` − `carwidth` − `enginelocation` − `horsepower`	0.9659	-215.6	After stepAIC
lmpg3	− `compressionratio` − `stroke`	0.9649	-211.1	Harmonise with lprice3

Table 3: Variable selection iteration process for $\log(\text{price})$.

Model	Regressor variables	R^2_{adj}	AIC	Remarks
lprice1	Selection from domain knowledge	0.9492	-266.7	Initial model
lprice2	− `cylinderNum` − `compressionratio` − `enginelocation` − `fuelsystem` − `stroke` − `horsepower`	0.9492	-273.5	After stepAIC
lprice3	+ `fuelsystem` − `carwidth`	0.9497	-271.0	Harmonise with lmpg3

2.2 Assumptions underlying the model

These models made a number of assumptions, including:

- Linearity and additivity, the assumption that $y = \log(\text{highwaympg}/\text{price})$ is a linear and additive function of the predictors.
- The model includes the relevant predictors and their functional forms are correct.
- The errors on the original highwaympg/price scale are multiplicative and follow a log-normal distribution.
- Homoscedasticity - the variance of the residuals is constant across all levels of the fitted values.
- The error for one observation (i.e., one car model) is uncorrelated with the error for any other car.
- The errors are uncorrelated with the predictor variables.
- There is no significant multicollinearity.
- There are no high influential points and the final model parameters aren't being disproportionately skewed by one or two extreme values.

- There is no sampling bias in our data.

3 Results

The two harmonised multiple linear regression models achieve a good fit with a R^2_{adj} of 0.965 and 0.950 for lmpg3 and lprice3 respectively (p <0.001 for both). Stepwise AIC produced parsimonious models and the regressors were aligned to ease with the interpretation of both models. The coefficients, standard errors, and p-values for both models are shown in Table 6.

The residual diagnostics shown in Figure 3 indicate that our model assumptions are broadly satisfied. On the log scale residuals are approximately normal with only minor variations at each extreme, the Residuals vs Fitted and Scale-Location figures indicate linearity, showing little to no evidence of heteroscedasticity. No observation has a Cook's distance greater or equal to 1, although some data points had a leverage of 1 (this is discussed further in our Limitations). There is no extreme collinearity with all VIF values being <4.3.

An advantage of the log transformation is that it makes the regression results easier to interpret by allowing coefficients to be read in percentage terms, where a one-unit change in a predictor corresponds to an approximate percentage change in the response variable. Mathematically, percentage effects can be transformed back from the log-scale coefficients as a coefficient β corresponds to a

$$100\,(\mathrm{e}^{\beta} - 1)$$

change in the outcome. This approach is more intuitive and meaningful for any analysis. Percentage effects for a selection of coefficients are shown in Table 4 for $\log(\text{highwaympg}/\text{price})$, $\log(\text{price})$, and $\log(\text{highwaympg})$ (inferred from the results of the other two).

Holding other features constant, cars with a turbocharger have a 19% lower mpg-per-dollar ratio and a 13% higher price (p <0.001 for both). Heavier and longer cars cost more and are less fuel-efficient. Adding 100lb of curb weight is associated with 7.2% decrease in mpg-per-dollar and a 5.2% increase in price (both p <0.001) and just a 1 inch increase in wheelbase is associated with a 1.8% decrease in mpg-per-dollar and a 1.6% increase in price.

Relative to convertibles, the reference body type, all other car types (hard tops, hatchbacks, sedans, and wagons) were between 17–21% cheaper and more efficient, achieving a 21–25% higher mpg-per-dollar (all $p \leq$ 0.012 and 0.010, respectively). Additionally, perhaps unsurprisingly, brand effects

Table 4: Predicted percentage change in a selection of key parameters from the fitted log-linear models. Entries equal $100\,(e^{\beta} - 1)$ and represent 'ceteris paribus' effects from including a feature or increasing a continuous variable by the increment shown (e.g., +1 inch of wheelbase, +1 cubic inch of engine size).

Predictor (increment)	Δ(mpg/price)	Δ(price)	Δ(mpg)
Having a turbocharged engine	-19.3%	$+13.0\%$	-8.9%
Having a rotory vs double overhead camshaft piston engine	-41.1%	$+30.5\%$	-23.5%
Wheelbase (+1 inch)	-1.8%	$+1.6\%$	-0.2%
Enginesize (+1 cubic inch)	-0.3%	$+0.2\%$	-0.1%
Car height (+1 inch)	$+2.2\%$	-2.9%	-0.7%

were substantial. BMW models are priced approximately 31% higher than Alfa-Romeos, the reference brand (p =0.004). A number of manufacturers including Chevrolet, Mitsubishi, Peugeot, Subaru, and Toyota exhibit a mpg-per-dollar ratio between 35–73% higher than Alfa-Romeos ($p \leq 0.02$).

4 Limitations

A key limitation in our data set is the curse of dimensionality: a symptom of which is data sparsity in a large feature space (Bellman, 1957; Keogh & Mueen, 2011; Niu et al., 2025). In our analysis this manifests through datapoints with a leverage of 1 within our models (Table 5). This limitation is due to categorical variables in our models with many levels (e.g., car manufacturer) combined with a small dataset. The curse of dimensionality is an established concept in statistics and machine learning, first noted by Bellman (1957).

Table 5: Observations for which the leverage equals 1 in either model. In each case a unique factor level or rare combination resulting in the fitted value equalling the observed value.

Datapoint	Explanation
19	The only Chevrolet car with a flathead engine in the dataset
30	The only Dodge car with multi-port fuel injection in the dataset
47	The only car with single-point field injection in the dataset in the dataset
59	The only car with both a rotary engine and multi-point field injection in the dataset
76	The only Mercury car in the dataset
126	The only Porsche with an overhead camshaft engine in the dataset
130	The only Porsche with a duel overhead camshaft engine in the dataset

Our VIF values and backward stepwise selection process confirmed the lack of any severe multi-collinearity in our models, but we know that some variables, such as wheelbase, engine size, and curbweight, are correlated. This is expected, a larger car is typically required to accommodate a larger engine, but this (non severe) collinearity increases the variance of our estimated coefficients.

The dataset omits `normalized-losses` and other information (i.e., trim levels, optional extras,

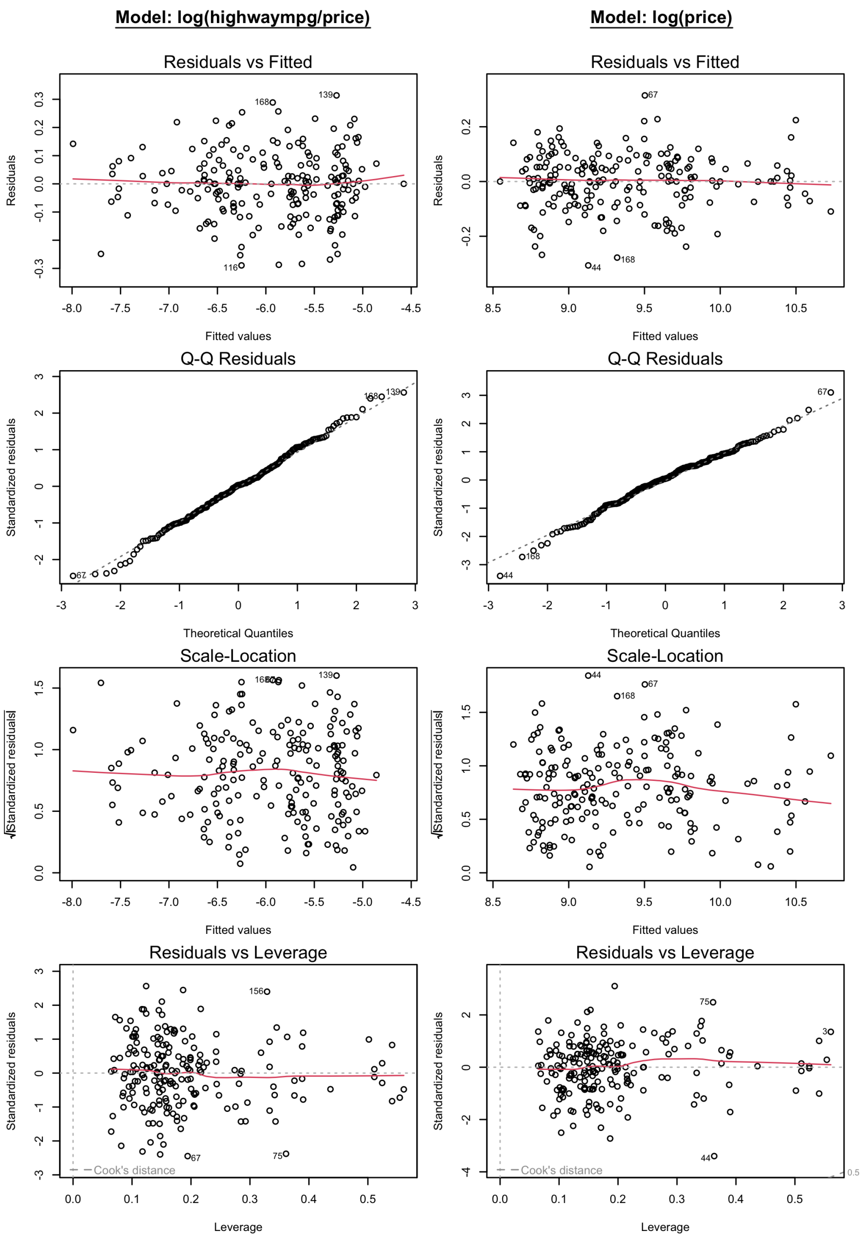

Figure 3: Side-by-side regression diagnostics for the final models $\log(\text{highwaympg}/\texttt{price})$ (left) and $\log(\text{price})$ (right), including residuals vs fitted, Normal Q-Q, scale-location, and residuals vs leverage.

maintenance costs) that would influence a 1985 car's cost. Whilst our model attempts to account for this through multiple regression analysis, any analysis is inherently limited to the variables that are included in the model. Additionally, the data are from 1985 and our findings can therefore only be interpreted in a historical context.

A final limitation is the observational nature of the dataset. As a result, we can only identify variables that are significantly *associated* with a car's price and fuel efficiency value. Since correlation does not imply causation and due to possible omitted variable bias (confounding), simultaneity and selection bias, we cannot infer causality from this analysis.

5 Future Improvements

We identified two broad areas for future improvements: the data (quality and quantity) and our modelling methods. For data, we would supplement the assignment dataset with the full 1985 Ward's Automotive Yearbook dataset (Schlimmer, 1985) to access `normalized-losses` and model the `symboling` risk. We would also enrich and modernise the dataset, adding model year, milage, trim, condition, and location, either from Kaggle used-car data or historical data from the 1980s-1990s. More data would enable the external validation of our models and having two time points would allow us to compare changes in the automotive industry over time.

For modelling methods, we would go beyond OLS regressions, using Generalised Linear Model (GLM) to model response variables having non-Gaussian error terms and adding shrinkage models (including ridge to address multicollinearity, lasso for variable selection, and elastic net) with cross-validation.

We would also consider modelling methods that do not assume linearity, such as tree-based models and neural networks, both of which would be particularly valuable for a price prediction model (although their improved accuracy is generally at the cost of interpretability). A critical area to pursue would be to draw causal inference, ideally by conducting a randomised experiment if possible, or by employing quasi-experimental techniques such as instrumental variables, difference-in-differences, regression continuity design and debiased machine learning methods. Combined with our data enrichments this would deliver a more robust analysis and build on the groundwork laid by this project.

References

Bellman, R. (1957). *Dynamic programming*. Princeton Univ. Pr.

Keogh, E., & Mueen, A. (2011). Curse of Dimensionality. In *Encyclopedia of Machine Learning* (pp. 257–258). Springer, Boston, MA. Available at: https://link.springer.com/rwe/10.1007/978-0-387-30164-8_192. (Accessed: October 29, 2025).

Niu, H., McCallum, G. B., Chang, A. B., Khan, K., & Azam, S. (2025). Exploring unsupervised feature extraction algorithms: Tackling high dimensionality in small datasets [Publisher: Nature Publishing Group]. *Scientific Reports*, *15*(1), 21973. Available at: https://www.nature.com/articles/s41598-025-07725-9. (Accessed: October 29, 2025).

Schlimmer, J. (1985). Automobile [Accessed: 2025-10-23]. Available at: https://doi.org/10.24432/C5B01C.

A GitHub Resources

All resources described in these appendices are available via a public Github:

https://github.com/ytterbiu/smm634-AS-g4-assignment/tree/main

An R Markdown Notebook is also available via GitHub pages for the above repository: https://ytterbiu.github.io/smm634-AS-g4-assignment/MSc_AS-SMM634-Group4-Project.html

B Supplementary Figures and Tables

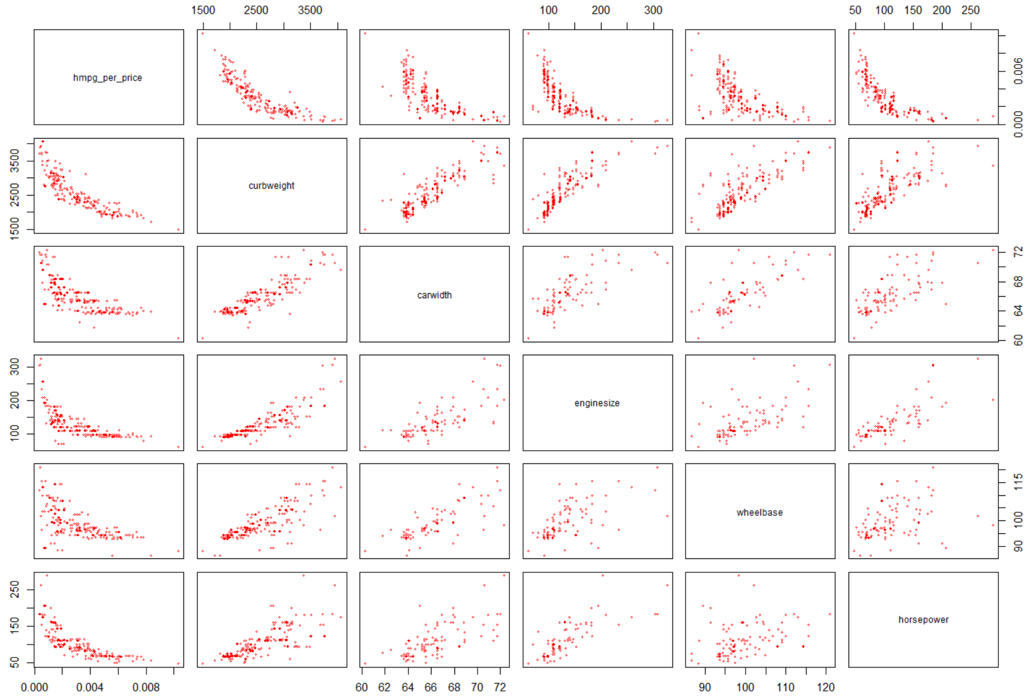

Figure 4: Pairwise scatterplots using the raw response `hmpg_per_price` and key predictors. Several relationships are curved with visible heteroscedasticity.

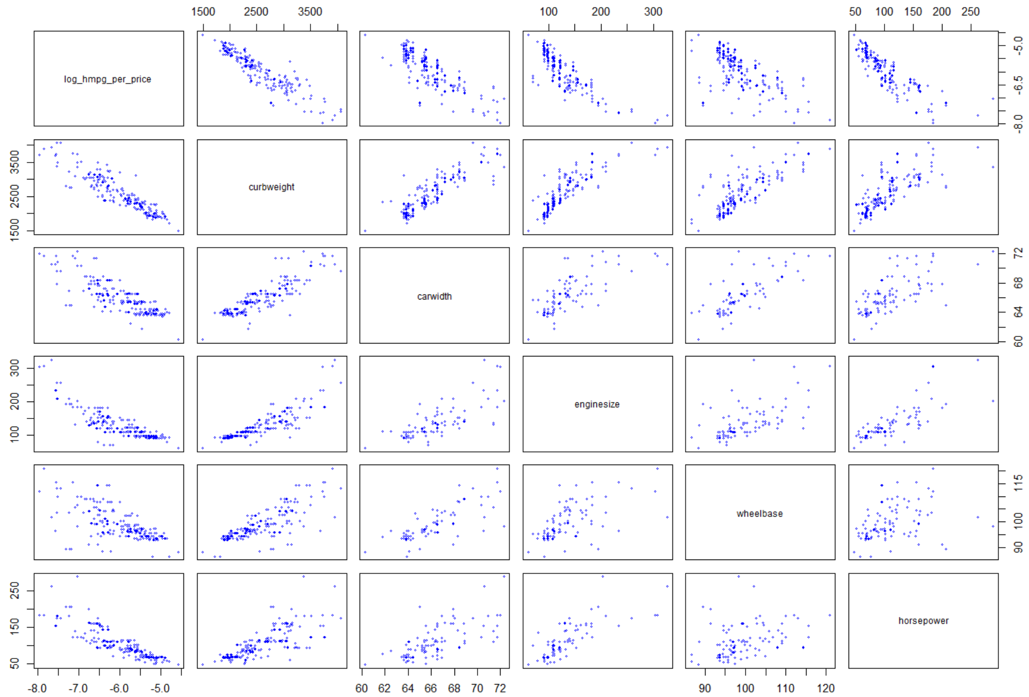

Figure 5: Pairwise scatterplots using the raw response `log_hmpg_per_price` and key predictors. Several relationships are curved with visible heteroscedasticity.

Table 6: Estimated coefficients ($\hat{\beta}$), standard errors, and p-values for the two fitted models: $\log(\texttt{highwaympg}/\texttt{price})$ and $\log(\texttt{price})$. Categorical predictors are dummy-coded with one reference level omitted.

	$\log(\texttt{highwaympg}/\texttt{price})$ model			$\log(\texttt{price})$ model		
Term	**Estimate**	**Std. Error**	**p-value**	**Estimate**	**Std. Error**	**p-value**
`(Intercept)`	-2.570	0.536	<0.001	7.740	0.463	<0.001
`aspirationturbo`	-0.215	0.039	<0.001	0.122	0.033	<0.001
`carbodyhardtop`	0.211	0.081	0.010	-0.192	0.070	0.007
`carbodyhatchback`	0.247	0.074	0.001	-0.214	0.064	0.001
`carbodysedan`	0.209	0.076	0.007	-0.166	0.066	0.012
`carbodywagon`	0.243	0.084	0.004	-0.195	0.072	0.008
`carheight`	0.022	0.009	0.012	-0.029	0.007	<0.001
`carManufactureraudi`	-0.068	0.124	0.584	0.057	0.107	0.594
`carManufacturerbmw`	-0.249	0.124	0.046	0.314	0.107	0.004
`carManufacturerbuick`	0.065	0.129	0.614	-0.002	0.111	0.985
`carManufacturerchevrolet`	0.469	0.140	0.001	-0.258	0.121	0.035
`carManufacturerdodge`	0.397	0.116	<0.001	-0.307	0.100	0.002
`carManufacturerhonda`	0.043	0.143	0.763	-0.032	0.123	0.793
`carManufacturerisuzu`	0.218	0.128	0.092	-0.109	0.111	0.326
`carManufacturerjaguar`	0.459	0.150	0.003	-0.296	0.130	0.024
`carManufacturermazda`	0.138	0.108	0.204	-0.080	0.093	0.395
`carManufacturermercury`	0.175	0.176	0.320	-0.103	0.152	0.500
`carManufacturermitsubishi`	0.463	0.117	<0.001	-0.366	0.101	<0.001
`carManufacturernissan`	0.271	0.104	0.010	-0.169	0.090	0.062
`carManufacturerpeugeot`	0.547	0.214	0.011	-0.522	0.185	0.005
`carManufacturerplymouth`	0.443	0.118	<0.001	-0.332	0.102	0.001
`carManufacturerporsche`	-0.266	0.174	0.129	0.331	0.150	0.029
`carManufacturerrenault`	0.379	0.140	0.008	-0.277	0.121	0.023
`carManufacturersaab`	0.049	0.117	0.676	0.051	0.101	0.617
`carManufacturersubaru`	0.586	0.202	0.004	-0.631	0.175	<0.001
`carManufacturertoyota`	0.301	0.098	0.002	-0.220	0.084	0.010
`carManufacturervolkswagen`	0.209	0.110	0.058	-0.127	0.095	0.183
`carManufacturervolvo`	0.232	0.123	0.060	-0.099	0.106	0.352
`curbweight`	-0.001	0.000	<0.001	0.001	0.000	<0.001
`enginetypedohcv`	0.394	0.193	0.043	-0.100	0.167	0.552
`enginetypel`	-0.128	0.181	0.480	0.194	0.156	0.216
`enginetypeohc`	0.022	0.061	0.712	-0.006	0.052	0.905
`enginetypeohcf`	-0.348	0.176	0.049	0.392	0.152	0.011
`enginetypeohcv`	-0.022	0.074	0.764	-0.003	0.064	0.969
`enginetyperotor`	-0.534	0.155	<0.001	0.266	0.134	0.049
`enginesize`	-0.003	0.001	0.002	0.002	0.001	0.048
`fuelsystem2bbl`	-0.239	0.105	0.024	0.130	0.090	0.153
`fuelsystem4bbl`	-0.183	0.186	0.325	0.007	0.160	0.964
`fuelsystemidi`	-0.015	0.117	0.901	0.136	0.101	0.180
`fuelsystemmfi`	-0.352	0.181	0.054	0.151	0.156	0.336
`fuelsystemmpfi`	-0.322	0.109	0.003	0.188	0.094	0.046
`fuelsystemspdi`	-0.375	0.126	0.003	0.196	0.109	0.074
`fuelsystemspfi`	-0.232	0.190	0.225	0.047	0.164	0.775
`peakrpm`	0.000	0.000	0.003	0.000	0.000	0.174
`wheelbase`	-0.018	0.005	<0.001	0.016	0.004	<0.001

C R Code

C.1 Extracting car manufacturer information

````
105 ```{r}
106 # extract car manufacturer from first part of CarName
107 df$carManufacturer <- sapply(strsplit(df$CarName, " +"), `[`, 1)
108 # print
109 table(df$carManufacturer)
110 ```
111
112 Our assumptions are as follows:
113
114 | Typo / Mis-spelling | Correct Spelling |
115 | ------------------- | ---------------- |
116 | maxda               | mazda            |
117 | Nissan              | nissan           |
118 | porcshce            | porsche          |
119 | toyouta             | toyota           |
120 | vokswagen           | volkswagen       |
121 | vw                  | volkswagen       |
122
123 ```{r}
124 df <- df %>% mutate(carManufacturer = case_match(carManufacturer,
125   "maxda" ~ "mazda",
126   "Nissan" ~ "nissan",
127   "porcshce" ~ "porsche",
128   "toyouta" ~ "toyota",
129   c("vokswagen", "vw") ~ "volkswagen",
130   .default = carManufacturer
131 ))
````

Listing C.1: Extracting the car manufacturer from the first part of `CarName`. We corrected for typos and errors using the mapping outlined in lines 113–120.

C.2 Defining and refining our models

````
```{r}

lmpg1 <- lm(formula = log(highwaympg/price) ~ aspiration +
       carbody +
````
````

```
        carheight +
        carManufacturer +
        compressionratio +
        curbweight +
        cylinderNum +
        carwidth +
        enginelocation +
        enginetype +
        enginesize +
        fuelsystem +
        peakrpm +
        stroke +
        wheelbase +
        horsepower,
      data = df)

summary(lmpg1)
print(AIC(lmpg1))
```
```

```{r}
stepAIC(
  lmpg1,
  scope = list(lower = ~ carManufacturer,
               upper = ~ .),
  direction = 'both',
  trace = 1
)
```

```{r}
lmpg2 <- lm(formula = log(highwaympg/price) ~ aspiration + carbody +
                  carheight +
        carManufacturer + compressionratio + curbweight + enginetype +
        enginesize + fuelsystem + peakrpm + stroke + wheelbase, data = df)

summary(lmpg2) #R^2: .9659
print(AIC(lmpg2))
```
```

~~~~

```{r}

lmpg3 <- lm(formula = log(highwaympg/price) ~ aspiration + carbody +
                wheelbase +
                carheight + curbweight +
                enginetype + enginesize +
                fuelsystem + peakrpm +
                carManufacturer, data = df)

summary(lmpg3) #R^2: .9649
print(AIC(lmpg3))
```


```{r}

lprice1 <- lm(formula = log(price) ~ aspiration + carbody + carheight +
       carManufacturer + compressionratio + curbweight + enginetype +
       enginesize + fuelsystem + peakrpm + stroke + wheelbase + horsepower,
     data = df)

summary(lprice1)
print(AIC(lprice1))
```


```{r}

stepAIC(
  lm(formula = log(price) ~ aspiration +
       carbody +
       carheight +
       carManufacturer +
       compressionratio +
       curbweight +
       cylinderNum +
       carwidth +
       enginelocation +
       enginetype +
~~~~

````
        enginesize +
        fuelsystem +
        peakrpm +
        stroke +
        wheelbase,
      data = df),
  scope = list(lower = ~ carManufacturer,
               upper = ~ .),
  direction = 'both',
  trace = 1
)
```


```{r}
lprice2 <- lm(formula = log(price) ~ aspiration + carbody + carheight +
       carManufacturer + curbweight + carwidth + enginetype + enginesize +
       peakrpm + wheelbase, data = df)

summary(lprice2)
print(AIC(lprice2))
```

```{r}

lprice3 <- lm(formula = log(price) ~ aspiration + carbody +
                wheelbase +
                carheight + curbweight +
                enginetype + enginesize +
                fuelsystem + peakrpm +
                carManufacturer, data = df)

summary(lprice3)
print(AIC(lprice3))
```
````

Listing C.2: Defining and improving our multiple linear regression models in R.
````